

- 9 (a) By reference to energy transfers, distinguish between electromotive force (e.m.f.) and potential difference (p.d.).

e.m.f.	Emf: electrical energy converted from other forms of energy per unit charge delivered round a complete circuit.	
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p.d.	Pd: The potential difference between two points in a circuit is the amount of energy converted from electrical to other forms of energy per unit charge passing from one point to the other	
.....		[2]

- (b) A circuit is set up as shown in Fig. 9.1.

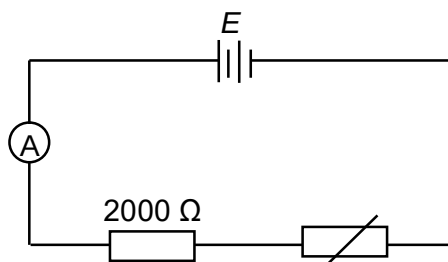


Fig. 9.1

The source of e.m.f. E is found to provide 2.4×10^5 J of electrical energy to the $2000\ \Omega$ resistor and thermistor when a charge of 2.2×10^4 C passes through the ammeter. At room temperature, the thermistor has a resistance of $1800\ \Omega$.

- (i) Sketch on Fig. 9.2 the variation with temperature θ of resistance R in a thermistor. [1]

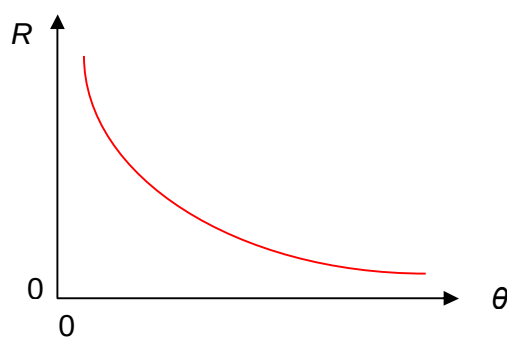


Fig. 9.2

- (ii) For the thermistor at room temperature,

1. show that E is 11 V. [1]

$$E = \frac{W}{Q} = \frac{2.4 \times 10^5}{2.2 \times 10^4} = 11\text{V} \quad [\text{M1}]$$

2. determine the time taken for a charge of $2.2 \times 10^4 \text{ C}$ to pass through the ammeter.

$$Q = It = \frac{E}{R}t$$

$$\Rightarrow t = \frac{QR}{E} = \frac{2.2 \times 10^4 \times 3800}{11} \quad [\text{M1}]$$

$$= 7.6 \times 10^6 \text{ s} \quad [\text{A1}]$$

time = s [2]

3. show that the fraction of power dissipated in the thermistor is 0.47. [1]

$$\text{fraction} = \frac{I^2 R_T}{I^2 R_T + I^2 R} = \frac{R_T}{R_T + R} = \frac{1800}{1800 + 2000} \quad [\text{M1}]$$

$$= 0.474$$

- (c) A uniform resistance wire PQ of length 1.2 m is subsequently connected across the resistor and thermistor, as shown in Fig. 9.3. A sensitive voltmeter is connected between point Y and a moveable contact M on the wire.

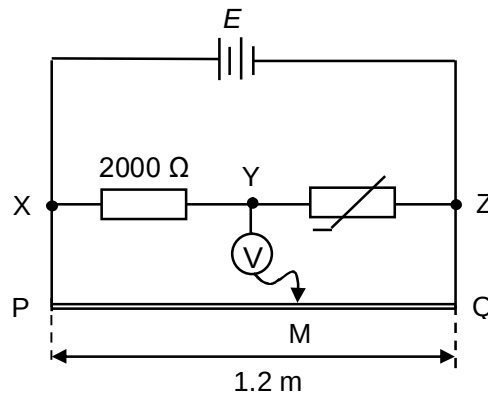


Fig. 9.3

- (i) At room temperature, the contact M is moved along PQ until the voltmeter shows zero reading.

Calculate the length of wire between M and Q.

Potential difference between MQ and YZ has to be the same for voltmeter to register zero reading.

$$\frac{L}{1.2} = \frac{1800}{3800} \quad [\text{M1}]$$

$$L = 0.568 \text{ m} \quad [\text{A1}]$$

length of wire = m [2]

- (ii) State and explain the effect, if any, on the length of the wire between M and Q for the voltmeter to remain at zero deflection if each of the following changes takes place independently.

1. The thermistor is warmed slightly.

(As temperature rises), resistance of thermistor decreases and by Potential Divider Principle, the potential difference across the thermistor drops.

For zero deflection, the potential difference across the thermistor and across MQ have to be the same. Since the pd across MQ is proportional to its length, MQ is shorter.

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.. [2]

2. A uniform wire with a bigger cross sectional area is used to replace PQ.

While the bigger cross-sectional area will result in a lower resistance in the wire PQ, as the potential difference across PQ did not change, the potential difference per unit length remains the same.

Since potential difference across thermistor (and resistor) did not change, the length of wire will not need to change.

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.. [2]

- (d) The circuit shown in Fig. 9.4 is used to compare potential differences.

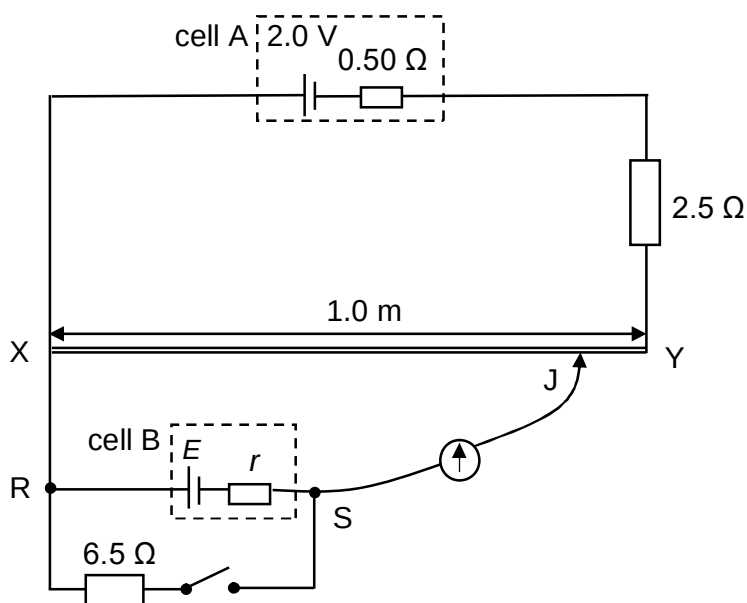


Fig. 9.4

The uniform resistance wire XY has length 1.0 m and resistance 8.0 Ω. Cell A has e.m.f. 2.0 V and internal resistance 0.50 Ω. Cell B has e.m.f. E and internal resistance r .

- (i) The switch is opened. The galvanometer shows no deflection when the moveable contact J is adjusted so that the length XJ is 0.90 m.

Show that the e.m.f. E of cell B is 1.3 V.

[2]

$$V_{XY} = \frac{8.0}{8.0 + 0.50 + 2.5} \times 2.0 = 1.455 \text{ V} \quad [\text{M1}]$$

At null deflection, $V_{XJ} = \text{e.m.f. of cell B.}$

$$E = \frac{0.90}{1.0} \times 1.455 = 1.31 \text{ V} \quad [\text{M1}]$$

(ii) The switch is now closed.

1. For the galvanometer to show no deflection, contact J has to be adjusted so that length XJ is 0.75 m.

Determine the internal resistance r of cell B.

At null deflection, $V_{XJ} = V_{RS}$

$$V_{XJ} = \frac{0.75}{1.0} \times 1.455 = 1.091 \text{ V} \quad [\text{M1}]$$

$$I = \frac{V}{R} = \frac{1.091}{6.5} = 0.1679 \text{ A} \quad [\text{M1}]$$

$$E = V + Ir \Rightarrow r = \frac{E - V}{I} = \frac{1.3 - 1.091}{0.1679} = 1.25 \text{ } \Omega \quad [\text{A1}]$$

$$r = \dots\dots\dots \Omega \quad [3]$$

2. A resistor is connected in parallel with the $6.5 \text{ } \Omega$ resistor.

Deduce how the balanced length XJ would be affected.

With resistor connected in parallel, load resistance of cell B would decrease.

This would cause terminal p.d. of cell B to decrease. [1]

Since no change in potential difference per unit length along wire XY,
Balance length XJ would decrease. [A1]

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..... [2]

END OF PAPER

